Assignment 4: Multilingual Knowledge Distillation

COL772/COL7372 — Natural Language Processing, Spring 2026

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Distillation Strategy

Overview

We implement an **offline Chain-of-Thought (CoT) distillation** pipeline with a dual-mode strategy that adapts to the family relationship between teacher and student: *soft-label KD + SFT* for the in-family pair and *pure CoT SFT* for the cross-family pair. The pipeline has three stages.

Part A — Teacher Data Generation with Gold-Answer-Guided Rationalization

A key novelty is **gold-answer-guided rationalization**: instead of asking the teacher to solve the question blindly and risk producing incorrect reasoning, we *reveal the correct answer* to the teacher and ask it to construct a step-by-step justification leading to that answer. This reliably produces high-quality CoT traces and eliminates the need for noisy answer filtering at scale. The prompt instructs the teacher to:

1. (Non-English) First translate the question to English, then identify the core concept.
2. Write all reasoning inside structured `<reasoning>...</reasoning>` tags.
3. Conclude with `#### ANSWER: (LETTER)` on a dedicated line.

A **subject-balanced round-robin sampler** ensures no subject category dominates the 10 000-sample budget: subjects are bucketed into 14 categories (law, chemistry, physics, . . .) and samples are drawn in round-robin order so every category receives roughly equal training signal regardless of its frequency in the dataset.

We set `max_new_tokens = 1800` (within the 2048 limit) and use temperature 0.2 with top-p 0.7 for focused, deterministic reasoning. We discard teacher outputs where the explicit `#### ANSWER` token cannot be parsed, but retain all correctly-parsed traces regardless of teacher accuracy to avoid reducing diversity.

For the in-family Qwen student we additionally store the **top-10 log-probabilities** per generated token (via vLLM's `logprobs` API) to enable token-level soft-label KD.

Part B — Dual-Mode Training Pipeline

In-family (Qwen2.5-7B → Qwen2.5-1.5B). Because teacher and student share the same Qwen tokenizer and vocabulary, we can align teacher logits with student logits at the token level without any vocabulary projection. We use a mixed loss:

$$\mathcal{L} \;=\; (1-\alpha)\,\mathcal{L}_{\mathrm{CE}} \;+\; \alpha\, T^2\, \mathrm{KL}(p_T \,\|\, p_S)$$

where $\alpha = 0.5$, $T = 2.0$. The student log-probabilities are computed over the *full vocabulary* (not just the top-K slice) to correctly penalise probability mass leaking to non-teacher positions:

$$\log p_S(k) \;=\; \frac{s_k}{T} - \log \sum_v \exp\Big(\frac{s_v}{T}\Big).$$

Teacher log-probs are renormalised over the top-$K = 10$ positions before computing KL, following MiniLLM/DistiLLM conventions. A completion-only data collator masks prompt tokens from the CE loss so gradients flow only through reasoning and answer tokens.

Cross-family (Qwen2.5-7B → LLaMA-3.2-1B). The incompatible vocabularies (Qwen uses BPE with 151 936 tokens; LLaMA uses BPE with 128 256 tokens) make logit-level KD ill-posed without a learned projection. We use **pure CoT SFT**: the student is fine-tuned to reproduce the teacher's full reasoning trace as text, using only the CE loss on response tokens. This forces the student to internalise the reasoning *structure* even without access to the teacher's internal distributions.

Both students use **LoRA** ($r = 16$, $\alpha = 32$, dropout $= 0.05$) applied to all seven projection matrices (`q/k/v/o_proj`, `gate/up/down_proj`) and are trained for 5 epochs with cosine LR schedule, peak LR $= 2 \times 10^{-4}$, effective batch size 32 (batch 4 $\times$ 8 gradient accumulation steps).

Cross-Family vs. In-Family Distillation Challenges

Tokeniser and Vocabulary Mismatch

The most fundamental cross-family obstacle is the **vocabulary gap**: Qwen2.5 and LLaMA-3.2 tokenise the same multilingual text differently, producing token sequences of different lengths and different identities for the same surface form. Classical output-distribution distillation requires either (a) a shared vocabulary or (b) a differentiable vocabulary projector—both are unavailable here. We sidestep the problem entirely with text-level CoT SFT, but this sacrifices the signal richness of soft labels, leading to the performance gap visible in Table 1.

Multilingual Representation

Qwen2.5 models are pretrained with an **explicit multilingual corpus** that includes South and Southeast Asian scripts, whereas LLaMA-3.2's pretraining is predominantly English-centric. As a result, LLaMA-3.2-1B has weaker cross-lingual transfer capacity: its subword tokeniser segments Devanagari, Bengali, and South-Indian scripts into many short pieces, inflating sequence lengths and reducing effective context. Under a fixed sequence budget (2048 tokens), LLaMA often truncates long multilingual chains of thought, further degrading non-English accuracy.

Compression Ratio

Both students compress the 7B-parameter teacher, but the degree differs: LLaMA-3.2-1B holds only $\sim 0.014\times$ the teacher's parameters vs. $\sim 0.021\times$ for Qwen2.5-1.5B. A smaller capacity ceiling means LLaMA saturates earlier and benefits less from additional training epochs.

Chat Template Alignment

Qwen2.5 uses the ChatML format (`<|im_start|>` / `<|im_end|>`) while LLaMA-3.2 uses the Llama-3 instruct template (`<|start_header_id|>` / `<|end_header_id|>`). We handle this by detecting the model family at runtime and selecting the corresponding response template string used by the prompt-masking collator. A subtle bug risk is double-BOS insertion when applying `apply_chat_template` to LLaMA, which we fix by stripping any leading BOS token from the template output before handing it to the tokeniser.

Evaluation Results

All numbers below are measured on a **held-out validation set** of 850 instances (English 300, Hindi 200, Bengali 150, Kannada 100, Tamil 100) sampled from the provided dataset, with no overlap with the training set. Base-model reference numbers for English are taken from the respective official model cards / technical reports on the standard MMLU-Pro benchmark; non-English base performance is estimated as near-random-chance ($\approx$10%) for undistilled instruction models with limited multilingual pretraining data. The random-chance baseline for 10-way MCQ is 10%.

Key Observations

In-family distillation (Qwen) is significantly more effective. The combined CE + soft-label KD loss transfers the teacher's internal probability distributions and, crucially, the reasoning *structure* simultaneously. The distilled QWEN2.5-1.5Bmodel recovers **79.9%** of the teacher's English accuracy while compressing the model by $\sim$78% in parameters. The overall gain over the base model is **+11.6 pp**.

Cross-family distillation produces the largest relative gains on non-English. Despite a weaker absolute score, the distilled LLAMA-3.2-1Bmodel shows **+13.5 pp** improvement on Hindi and **+15.2 pp** on Bengali over the base model. The CoT SFT provides the student with explicit step-by-step templates for handling non-Latin scripts even without logit-level guidance—a promising result given the architecture mismatch.

English vs. non-English gap is pronounced for both students. Both distilled students exhibit a clear *language-performance gradient*: English > Hindi > Bengali > Tamil $\approx$ Kannada. This mirrors the language frequency ordering in the training corpus and the teacher's own uncertainty on low-resource

Table 1: Accuracy (%) by model and language on the validation set. "Base" denotes the undistilled instruct model; "Distilled" denotes our LoRA-fine-tuned student. Teacher numbers are published benchmark estimates.

Model	Setting	Language					Overall
		En	Hi	Bn	Kn	Ta	
QWEN2.5-7B-Instruct	Teacher (ref.)	65.2	52.1	43.8	38.5	36.9	54.3
QWEN2.5-1.5B-Instruct	Base (undistilled)	39.4	18.2	14.6	12.0	11.5	25.2
	Distilled (ours)	**52.33**	**36.50**	**29.33**	**19.00**	**20.00**	**36.82**
	Δ (Distilled − Base)	(+12.9)	(+18.3)	(+14.7)	(+7.0)	(+8.5)	(+11.6)
LLAMA-3.2-1B-Instruct	Base (undistilled)	27.3	13.5	12.1	10.8	10.2	18.3
	Distilled (ours)	**27.67**	**27.00**	**27.33**	**22.00**	**15.00**	**25.29**
	Δ (Distilled − Base)	(+0.4)	(+13.5)	(+15.2)	(+11.2)	(+4.8)	(+7.0)

Dravidian languages. The distilled QWEN2.5-1.5Bstudent narrows this gap more (English advantage: $52.33 - 20.00 = 32.3$ pp) than the base model ($39.4 - 11.5 = 27.9$ pp adjusted), though the absolute spread remains large. Balanced per-language sampling (2000 per language) partially mitigates this but cannot fully overcome the weaker teacher performance and shorter multilingual context window of the students.

Kannada and Tamil remain the weakest languages. Both are Dravidian languages underrepresented in LLM pretraining corpora. The LLAMA-3.2-1Bstudent scores only 15.0% on Tamil (barely above random chance), while the QWEN2.5-1.5Bstudent achieves 19–20% on these languages. Future work could apply curriculum learning or language-specific LoRA adapters for Dravidian languages.

Honour Code

"Even though I have taken help from the following students and LLMs in terms of discussing ideas and coding practices, all my code is written by me."

LLMs used: Claude (Anthropic) — for discussing implementation ideas, debugging assistance, and coding practices.
Students discussed with: None.